

■ European Football Analytics

2025–2026 Season · Top 5 Leagues · Player Statistics

Data Science Internship Project | Decode Labs

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github.com/mohamedmamdouh342-hash/decode-labs-data-science-internship

#	Task	Status
1	Data Collection & Dataset Understanding	■ Complete
2	Data Cleaning & Preprocessing	■ Complete
3	Exploratory Data Analysis (EDA)	■ Complete
4	Data Visualization — 20 Charts	■ Complete
5	Predictive Modeling — 5 Models	■ Complete

Library	Purpose
pandas 2.0+	Data loading, cleaning, and analysis
numpy 1.24+	Numerical operations
matplotlib 3.7+	Chart drawing and customization
seaborn 0.12+	Statistical visualizations (heatmap, boxplot)
scikit-learn 1.3+	Machine learning models and evaluation
jupyter 1.0+	Interactive notebook environment

1. Dataset Overview

The dataset is the **European Football Players Statistics 2025–2026**, sourced from FBref (Football Reference). It covers player performance across the top five European leagues, providing attacking, defensive, and goalkeeping contributions at the player level.

Total Players (Raw)	After Cleaning	Total Columns (Raw)	Leagues
2,839	2,363	102	5

Column Categories

Category	Cols	Columns Included
Player Identity	7	Player, Nation, Pos, Squad, Comp, Age, Born
Playing Time	4	MP, Starts, Min, 90s
Attacking	7	Gls, Ast, G+A, G-PK, PK, PKatt, G+A-PK
Shooting	7	Sh, SoT, SoT%, Sh/90, SoT/90, G/Sh, G/SoT
Discipline	6	CrdY, CrdR, 2CrdY, Fls, Fld, OG
Defending	3	TklW, Int, Crs
Goalkeeping	13	GA, GA90, SoTA, Saves, Save%, W, D, L, CS, CS%, PKA, PKsv, PKm
Playing Time Adv	13	Mn/MP, Min%, Mn/Start, Compl, Subs, Mn/Sub, unSub, PPM, onG, onGA, +/-, +/-90, On-Off

Players per League

League	Players	Total Goals	Avg Goals / Player	Avg Assists / Player
Premier League	457	981	2.147	1.486
La Liga	510	984	1.929	1.353
Bundesliga	429	965	2.249	1.615
Serie A	508	889	1.750	1.177
Ligue 1	459	827	1.802	1.268

■ Column Glossary — Every Stat Explained

Every column in the dataset explained in plain English — what it means in football, why it matters, and which positions it applies to.

■ Player Identity

Column	Full Name	What it means in football	For
Player	Player Name	Full name of the player	All
Nation	Nationality	Country the player represents internationally	All
Pos	Position Code	GK=Goalkeeper, DF=Defender, MF=Midfielder, FW=Forward. Can be multi-coded e.g. MF,FW means primarily a midfielder who also plays forward	All
Squad	Club Team	The club the player plays for this season	All
Comp	Competition/League	League the player competes in (e.g. Premier League, La Liga)	All
Age	Age	Player age in years at the time of the data snapshot	All
Born	Birth Year	Year the player was born	All

■■ Playing Time

Column	Full Name	What it means in football	For
MP	Matches Played	Total league matches the player appeared in this season	All
Starts	Starts	Matches where the player started in the first 11	All
Min	Minutes Played	Total minutes played across all appearances this season	All
90s	90-Minute Units	Minutes played ÷ 90. A player with 900 mins has 10.0 "90s". Used to normalize stats — a 5-game player can be fairly compared to a 30-game player	All

■ Attacking

Column	Full Name	What it means in football	For
Gls	Goals	Total goals scored this season (includes penalties)	FW/MF
Ast	Assists	Last pass before a goal was scored	FW/MF
G+A	Goals + Assists	Combined goal contributions	FW/MF
G-PK	Non-Penalty Goals	Goals scored NOT from the penalty spot — shows open-play finishing quality without the "easy" penalty goals	FW/MF
PK	Penalty Goals	Goals scored from the penalty spot only	FW
PKatt	Penalty Attempts	Number of penalties the player was awarded and took	FW
G+A-PK	Non-Penalty G+A	Goals + Assists with penalty goals removed — the purest measure of open-play contribution	FW/MF

■ Shooting

Column	Full Name	What it means in football	For
Sh	Shots	Total shots attempted including off-target and blocked	FW/MF
SoT	Shots on Target	Shots that forced a save or scored — blocked shots by outfield players NOT counted	FW/MF
SoT%	Shot Accuracy %	$\text{SoT} \div \text{Sh} \times 100$. How accurate the player shoots. World class = 50%+	FW/MF
Sh/90	Shots per 90	Average shots per 90 minutes — removes playing time bias	FW/MF
SoT/90	Shots on Target per 90	Average shots on target per 90 minutes	FW/MF
G/Sh	Goals per Shot	$\text{Goals} \div \text{total shots}$. Direct finishing quality measure. Average = 0.10, elite = 0.20+	FW
G/SoT	Goals per Shot on Target	$\text{Goals} \div \text{shots on target}$. Conversion of clear chances. 0.30+ = world-class	FW

■ Discipline

Column	Full Name	What it means in football	For
CrdY	Yellow Cards	Total yellow cards received this season	All
CrdR	Red Cards	Total red cards received (direct sending-off)	All
2CrdY	Double Yellow	Times the player got two yellows in one match, resulting in a red	All
FIs	Fouls Committed	Fouls committed against opponents — high for aggressive pressers	DF/MF
FId	Fouls Drawn	Fouls won — how often opponents foul them. High = difficult to defend	FW/MF
OG	Own Goals	Goals accidentally scored into own net	All

■■ Defending

Column	Full Name	What it means in football	For
TkIW	Tackles Won	Tackles where the player successfully won the ball — failed tackles not counted	DF/MF
Int	Interceptions	Times the player cut out a pass intended for an opponent — reading the game to intercept before it reaches the target	DF/MF
Crs	Crosses	Total crosses delivered into the penalty area from wide positions — relevant for wingers and full-backs	MF/DF

■ Goalkeeping

Column	Full Name	What it means in football	For
GA	Goals Against	Total goals the goalkeeper conceded this season	GK
GA90	Goals Against per 90	Average goals conceded per 90 minutes. Lower is better	GK
SoTA	Shots on Target Against	Total shots on target faced — reflects the GK's workload	GK
Saves	Saves	Total saves made. $\text{Saves} = \text{SoTA} - \text{GA}$	GK
Save%	Save Percentage	$\text{Saves} \div \text{SoTA} \times 100$. Primary GK quality measure. 70% = solid, 75% = excellent, 80% = elite	GK
W	Wins	Matches won by the team while the GK was playing	GK
D	Draws	Matches drawn while the GK was playing	GK
L	Losses	Matches lost while the GK was playing	GK
CS	Clean Sheets	Matches where the team conceded zero goals — most prestigious GK achievement	GK
CS%	Clean Sheet %	Percentage of matches played that ended with a clean sheet	GK
PKA	Penalty Kicks Allowed	Penalties that were scored against the goalkeeper	GK
PKsv	Penalty Kicks Saved	Penalties the goalkeeper stopped	GK
PKm	Penalty Kicks Missed	Penalties the opponent missed — not saved, but still not conceded	GK

■ Advanced Playing Time

Column	Full Name	What it means in football	For
Mn/MP	Minutes per Match	Average minutes per appearance. 90 = full match. 45 = half a match on average	All
Min%	Minutes Percentage	Percentage of all possible league minutes the player was on pitch. 100% = every minute of every match	All
Mn/Start	Minutes per Start	Average minutes per match when starting. Below 90 = regularly substituted off	All
Compl	Complete Matches	Matches where the player played all 90 minutes without being substituted	All
Subs	Substitute Appearances	Matches where the player came on as a substitute	All
Mn/Sub	Minutes per Sub	Average minutes played when coming off the bench	All
unSub	Unused Substitutes	Times the player was named on the bench but never brought on	All
PPM	Points per Match	Average league points (3 win / 1 draw / 0 loss) earned by the team in matches this player appeared in — reflects team quality around them	All
onG	Goals Scored On-Pitch	Total goals scored BY the team while this player was on the field	All

Column	Full Name	What it means in football	For
onGA	Goals Conceded On-Pitch	Total goals conceded BY the team while this player was on the field. High onGA for a DF = poor defensive contribution	All
+/-	Plus / Minus	Goals scored by team MINUS goals conceded while this player was on pitch. Positive = team outscored opponents during their minutes. Borrowed from ice hockey analytics	All
+/-90	Plus/Minus per 90	+/- divided by 90-minute units — normalizes the value for fair comparison between players with different playing time	All
On-Off	On vs Off Differential	Compares team goal difference WITH this player on pitch vs WITHOUT them. Positive = team performs better with this player. The truest measure of actual impact on team results	All

■ 2. Data Cleaning & Preprocessing

The raw dataset required systematic cleaning before analysis. Eight steps were applied to ensure data quality, consistency, and analysis readiness.

Step	Action	Result
1	Drop redundant merged-section columns (repeated across 5 FBref tables)	102 → 66 remaining columns
2	Remove duplicate transfers — keep entry with highest minutes	2,839 → 2,687 players
3	Filter players with fewer than 90 minutes played	2,687 → 2,363 players
4	Standardize positions: MF,FW → MF as primary role	4 clean roles: GK / DF / MF / FW
5	Clean league names — remove country prefixes	Premier League, La Liga, etc.
6	Fix data types — Age and Born converted to numeric	Correct numeric analysis
7	Fill missing values — rate stats → 0, small gaps → median	0 remaining NaN cells
8	Feature engineering — per-90 metrics, age groups, composite scores	10 new analytical columns

Position Distribution After Cleaning

Position	Count	Percentage
Midfielder (MF)	1,057	44.7%
Defender (DF)	733	31.0%
Forward (FW)	386	16.3%
Goalkeeper (GK)	187	7.9%

Engineered Features

Feature	Formula / Definition	Purpose
Gls_p90	Goals ÷ (Min÷90)	Goals per 90 mins — normalizes for playing time
Ast_p90	Assists ÷ (Min÷90)	Assists per 90 mins
GA_p90	(G+A) ÷ (Min÷90)	Goal contributions per 90 mins
TklW_p90	TklW ÷ (Min÷90)	Defensive tackles per 90 mins
Int_p90	Int ÷ (Min÷90)	Interceptions per 90 mins
AgeGroup	Age brackets	U20, 20-22, 23-25, 26-28, 29-31, 32-34, 35+
DefScore	TklW + Int	Combined defensive contribution metric
AttScore	Gls×4 + Ast×2 + SoT×0.5 + Sh×0.2	Weighted attacking contribution

Feature	Formula / Definition	Purpose
MidScore	$Gls \times 2.5 + Ast \times 2.5 + TklW \times 0.8 + Int \times 0.8$	Balanced midfield contribution
PerfScore	Position-specific weighted formula	Unique score per position (see Task 5)

■ 3. Exploratory Data Analysis

3.1 Descriptive Statistics

Stat	Mean	Median	Max	Min	Std
Age	26.20	26.00	42.00	16.00	4.47
Minutes	1,441	1,360	3,420	90	898
Goals	1.97	1.00	36.00	0.00	3.12
Assists	1.37	1.00	21.00	0.00	1.93
Shots	18.24	12.00	146.00	0.00	19.45
SoT	6.27	4.00	67.00	0.00	7.91
TkIW	14.12	11.00	72.00	0.00	12.49
Int	12.02	8.00	77.00	0.00	12.05

■ Median goals = 1 while Mean = 1.97 — a right-skewed distribution where a small elite of ~30 players pull the average up. Over 40% of outfield players score 0 or 1 goal all season.

3.2 Goals by League

League	Players	Total Goals	Avg Goals / Player	Avg Assists / Player
Bundesliga	429	965	2.249 ← Most attacking	1.615
Premier League	457	981	2.147	1.486
La Liga	510	984	1.929	1.353
Ligue 1	459	827	1.802	1.268
Serie A	508	889	1.750 ← Most defensive	1.177

3.3 Average Stats by Position

Position	Goals	Assists	Shots	SoT	TkIW	Int	Crosses
Forward (FW)	4.88	1.54	33.52	13.50	6.59	3.52	14.74
Midfielder (MF)	2.09	1.89	21.60	7.11	16.42	12.33	37.67
Defender (DF)	0.76	0.87	9.98	2.86	18.28	19.08	20.41
Goalkeeper (GK)	0.00	0.08	0.05	0.01	0.29	0.24	0.01

■ *Positional DNA is completely distinct. Forwards dominate goals/shots. Midfielders lead in crosses. Defenders top tackles and interceptions. Goalkeepers show near-zero across all outfield metrics — making them the easiest position to identify by machine learning.*

3.4 Age and Performance

Age Group	Players	Avg Goals	Avg Goals/90	Avg Assists	Avg TklW
U20	216	1.361	0.161	0.875	8.532
20–22	480	1.798	0.121	1.402	14.612
23–25	556	2.390	0.133	1.622	17.081
26–28	482	2.427	0.143	1.618	16.824
29–31	264	2.402	0.141	1.697	16.087
32–34	135	1.970	0.116	1.333	15.741
35+	43	2.093	0.167	1.279	10.884

■ *Performance peaks at 26–31 years (avg 2.40–2.43 goals per player). Under-22 players average 1.36–1.80 goals — confirming they are still in development.*

3.5 Top Performers — Attacking

Top 5 Goal Scorers

Player	Club	League	Goals	Shots	SoT	G/Sh h
Harry Kane	Bayern Munich	Bundesliga	36	119	67	0.22
Erling Haaland	Man City	Premier League	27	126	59	0.19
Kylian Mbappé	Real Madrid	La Liga	25	146	63	0.12
Vedat Muriqi	Mallorca	La Liga	23	114	50	0.16
Igor Thiago	Brentford	Premier League	22	84	43	0.17

Top 5 Assist Providers

Player	Club	League	Assists	Goals	Crosses
Bruno Fernandes	Manchester Utd	Premier League	21	9	181
Michael Olise	Bayern Munich	Bundesliga	19	15	146
Federico Dimarco	Inter	Serie A	16	7	295
Julian Ryerson	Dortmund	Bundesliga	15	0	250
Luis Díaz	Bayern Munich	Bundesliga	14	15	45

Most Clinical Finishers (min 10 shots — G/Sh)

Player	Club	League	Goals	Shots	G/Sh
Raphaël Guerreiro	Bayern Munich	Bundesliga	5	12	0.42
Sebastian Nanasi	Strasbourg	Ligue 1	7	19	0.37
Orri Steinn Óskarsson	Real Sociedad	La Liga	9	27	0.33
Santiago Bueno	Wolves	Premier League	3	10	0.30
Aleksandar Pavlovic	Bayern Munich	Bundesliga	3	10	0.30

■ *G/Sh reveals finishing quality independently of playing time. Guerreiro converted 42% of shots — elite level. Average across all players is ~0.10 (1 goal every 10 shots).*

3.6 Top Defensive Players

Player	Club	League	Pos	TkIW	Int	Total
James Garner	Everton	Premier League	MF	72	57	129
Victor Nelsson	Hellas Verona	Serie A	DF	45	75	120
Malang Sarr	Lens	Ligue 1	DF	44	76	120

Player	Club	League	Pos	TklW	Int	Total
Jon Aramburu	Real Sociedad	La Liga	DF	66	47	113
Morten Frendrup	Genoa	Serie A	MF	71	41	112
Senne Lynen	Werder Bremen	Bundesliga	MF	33	77	110

3.7 Top Goalkeepers by Save%

Player	Club	League	GA	Saves	Save%	Clean Sheets
Joan García	Barcelona	La Liga	21	74	80.4%	15
Ivan Provedel	Lazio	Serie A	27	87	78.6%	12
Hervé Koffi	Angers	Ligue 1	36	120	78.3%	10
Jean Butez	Como	Serie A	29	86	78.2%	19
Mile Svilar	Roma	Serie A	31	107	78.1%	18
Mike Maignan	Milan	Serie A	35	106	77.6%	14

■ 4. Data Visualization — 20 Charts

20 professional charts produced using matplotlib and seaborn. Charts 1–10 cover standard football analysis with league and position color coding. Charts 11–20 reveal advanced hidden insights using combinations of variables not visible in simple summaries. All charts are saved to the visuals/ folder at 150 DPI.

#	Chart	Football Question Answered
1	■ Top 15 Goal Scorers	Who scored the most goals this season?
2	■ Top 15 Assist Providers	Who created the most for teammates?
3	■ Goals Distribution by League	How are goals distributed — which leagues have depth?
4	■ Player Position Donut	What is the composition of players by role?
5	■ Age Distribution	Where is the prime age zone across positions?
6	■ Goals vs Assists Scatter	Who scores AND creates? Who specializes?
7	■ Correlation Heatmap	Which statistics are linked to each other?
8	■ Normalized Stats by Position	How does each position differ across all metrics?
9	■■ Top 12 Defensive Players	Who dominated defensively this season?
10	■ Top 10 Goalkeepers by Save%	Who stopped the highest percentage of shots?
11	■ Shooting Efficiency Quadrant	ADVANCED: Who is both accurate AND clinical? The elite vs the wasteful
12	■ Goal-Scoring Aging Curve	ADVANCED: At what exact age does each position peak?
13	■ League DNA Radar	ADVANCED: What is the statistical fingerprint of each league?
14	■■ Player Value Map	ADVANCED: Who produces at high rate AND high volume?
15	■■ Defensive Workrate Heatmap	ADVANCED: Which leagues demand the most defensive work per position?
16	■ Underplayed Gems	ADVANCED: Which forwards have elite per-90 rates but low minutes?
17	■ Discipline Analysis	ADVANCED: Cards by league and position — who is most aggressive?
18	■ Goals vs +/- Team Impact	ADVANCED: Do top scorers actually help their teams win?
19	■ GK Workload vs Performance Matrix	ADVANCED: Who faces the most shots AND saves the most?
20	■ Performance Score Rankings	ADVANCED: Who are the best players at their specific job?

■ 5. Predictive Models

Two machine learning approaches: a position role classifier (Part A) and four position-specific regression models (Part B). A single universal model is misleading — a defender with 0 goals is performing perfectly, but a global goals model would penalize them. Separate models respect what each position is actually supposed to do.

Part A — Role Classifier: Predict Position from Statistics

A Random Forest Classifier trained to predict a player's position (FW / MF / DF / GK) using only season statistics — no position label used as input. Mirrors how real scouting databases auto-categorize players from raw performance data.

Parameter	Value
Algorithm	Random Forest Classifier
Number of Trees	200
Max Tree Depth	12
Min Samples Leaf	3
Class Weighting	Balanced — handles GK class imbalance
Train / Test Split	80% / 20% (1,890 train / 473 test)
Feature Scaling	StandardScaler
Features Used	Gls, Ast, G+A, Sh, SoT, TklW, Int, Crs, GA, Saves, Save%, CS, Min, Age
Overall Accuracy	76.11%

Classification Results by Position

Position	Precision	Recall	F1-Score	Support	Note
DF (Defender)	72%	78%	75%	147	Main confusion with MF
FW (Forward)	73%	79%	76%	77	Some confusion with MF
GK (Goalkeeper)	100%	100%	100%	37	■ Perfect — unique stats
MF (Midfielder)	76%	69%	73%	212	Main confusion with DF

■ *Goalkeepers: 100% accuracy — GA, Saves, Save%, CS are uniquely non-zero only for the 187 GKs. Main confusion is DF vs MF (33 DFs classified as MF) — this reflects a real football reality: defensive midfielders share the same tackle/interception profile as central defenders. Even coaches debate where some players belong.*

Feature Importance — Role Classifier

Rank	Feature	Importance	Football Meaning
1	Int (Interceptions)	14.09%	Separates DF from FW/MF

Rank	Feature	Importance	Football Meaning
2	Sh (Shots)	10.87%	Separates FW from DF/GK
3	Saves	10.37%	Perfect GK identifier
4	Save%	10.37%	Perfect GK identifier
5	Crs (Crosses)	9.81%	Differentiates MF from FW
6	TkIW (Tackles Won)	8.67%	Separates DF/MF from FW
7	SoT (Shots on Target)	7.59%	Separates FW from others
8	Min (Minutes)	6.09%	Playing time context

Part B — Position-Specific Regression Models

Four regression models — one per position — each predicting the metric that defines success for that role. All four algorithms compared using 5-fold cross-validation.

■ Forwards Model — Target: Goals Scored (Gls)

Why this target: A forward's primary job is to score. Goals is the most direct measure of whether they are doing their job.

Features: Sh, SoT, SoT%, Sh/90, SoT/90, G/Sh, Min, 90s, Age, Off, +/-, PPM

Model	MAE	RMSE	R ²	CV-R ²
Linear Reg	1.262	1.683	0.914	-0.173
Ridge	1.250	1.684	0.914	-0.077
Random Forest	1.011	1.480	0.934	0.685
Grad Boost	1.048	1.671	0.915	0.689

■ Midfielders Model — Target: Assists (Ast)

Why this target: A midfielder's defining creative contribution — how well they connect play to goal-scoring opportunities.

Features: Crs, Sh, SoT, Min, 90s, Age, Fls, Int, TkIW, +/-, PPM, onG, GlS

Model	MAE	RMSE	R ²	CV-R ²
Linear Reg	1.043	1.401	0.512	0.273
Ridge	1.043	1.399	0.514	0.277
Random Forest	1.103	1.452	0.477	0.048
Grad Boost	1.168	1.535	0.415	0.040

■■ Defenders Model — Target: Defensive Score (TkIW + Int)

Why this target: A defender's job is to win the ball and cut out passes. DefScore combines the two purest defensive actions.

Features: Min, 90s, Age, CrdY, Fld, Crs, +/-, PPM, onGA, Starts, Fls

Model	MAE	RMSE	R ²	CV-R ²
Linear Reg	7.759	10.946	0.777	0.333
Ridge	7.756	10.929	0.777	0.334
Random Forest	7.677	11.094	0.771	-0.142
Grad Boost	8.063	11.418	0.757	-0.441

■ Goalkeepers Model — Target: Save Percentage (Save%)

Why this target: The primary measure of GK shot-stopping quality, independent of how many shots they face.

Features: SoTA, CS, W, D, L, Min, 90s, Age, GA, GA90, PPM

Model	MAE	RMSE	R ²	CV-R ²
Linear Reg	4.024	5.804	0.605	0.390
Ridge	3.828	5.709	0.618	0.573
Random Forest	3.871	5.085	0.697	0.365
Grad Boost	3.606	4.609	0.751	0.286

Model Performance Summary

Model	Target	Best Algorithm	Score	Metric
Role Classifier	Position (GK/DF/MF/FW)	Random Forest	76.11%	Accuracy
Forward Model	Goals	Random Forest	0.934	R ²
Midfielder Model	Assists	Ridge	0.514	R ²
Defender Model	Defensive Score	Ridge	0.777	R ²
Goalkeeper Model	Save%	Grad Boost	0.751	R ²

■ Performance Score Rankings — 2025/26

Custom weighted scoring formula per position — each position scored based on what their role specifically demands in football. FW: $Gls \times 4 + Ast \times 2 + SoT \times 0.5 + Sh \times 0.2$ | MF: $Gls \times 2.5 + Ast \times 2.5 + TklW \times 0.8 + Int \times 0.8$ | DF: $TklW \times 2.5 + Int \times 2.5 + GlS \times 1.5 + Ast \times 1.0$ | GK: $Saves \times 1.5 + Save\% \times 0.3 + CS \times 2.0$

■ Top 10 Forwards

Player	Club	League	Age	Score
Harry Kane	Bayern Munich	Bundesliga	32	211.3
Erling Haaland	Man City	Premier League	25	178.7
Kylian Mbappé	Real Madrid	La Liga	27	170.7
Vedat Muriqi	Mallorca	La Liga	32	141.8
Deniz Undav	Stuttgart	Bundesliga	29	137.8
Igor Thiago	Brentford	Premier League	24	128.3
Esteban Lepaul	Rennes	Ligue 1	26	125.3
Lautaro Martínez	Inter	Serie A	28	117.9
Vinicius Júnior	Real Madrid	La Liga	25	117.0
Ante Budimir	Osasuna	La Liga	34	109.1

■ Top 10 Midfielders

Player	Club	League	Age	Score
James Garner	Everton	Premier League	25	125.7
Bruno Fernandes	Manchester Utd	Premier League	31	119.8
Nicolás Paz	Como	Serie A	21	111.4
Federico Dimarco	Inter	Serie A	28	106.3
Michael Olise	Bayern Munich	Bundesliga	24	105.0
Christoph Baumgartner	RB Leipzig	Bundesliga	26	102.1
Elliot Anderson	Nott. Forest	Premier League	23	100.8
Mateus Fernandes	West Ham	Premier League	21	99.9
Luis Díaz	Bayern Munich	Bundesliga	29	96.5
Moisés Caicedo	Chelsea	Premier League	24	94.8

■ Top 10 Defenders

Player	Club	League	Age	Score
Malang Sarr	Lens	Ligue 1	27	301.0
Victor Nelsson	Hellas Verona	Serie A	27	300.0
Jon Aramburu	Real Sociedad	La Liga	23	284.0
Neco Williams	Nott. Forest	Premier League	25	273.5
Oumar Solet	Udinese	Serie A	26	265.5
Carlos Romero	Espanyol	La Liga	24	257.0
A. Caracciolo	Pisa	Serie A	35	255.5
Iglesias	Getafe	La Liga	27	253.5
D. Affengruber	Elche	La Liga	25	252.5
Adrien Truffert	Bournemouth	Premier League	24	241.5

■ Top 10 Goalkeepers

Player	Club	League	Age	Score
Aarón Escandell	Oviedo	La Liga	30	259.6
Moritz Nicolas	Gladbach	Bundesliga	28	252.2
Marko Dmitrovi■	Espanyol	La Liga	34	241.4
Emil Audero	Cremonese	Serie A	29	228.3
Mathew Ryan	Levante	La Liga	34	225.0
Hervé Koffi	Angers	Ligue 1	29	223.5
David de Gea	Fiorentina	Serie A	35	220.4
Mile Svilar	Roma	Serie A	26	219.9
David Soria	Getafe	La Liga	33	219.2
Elia Caprile	Cagliari	Serie A	24	218.7

■ 6. Key Findings & Conclusion

Goal-scoring is extremely rare

Median goals = 1 across all outfield players. Over 40% score 0 or 1 goal all season. Only ~30 players reach double figures. Goal-scoring is a rare, elite specialization.

Harry Kane leads Europe with 36 goals

Kane tops all scorers with 36 goals in the Bundesliga — 9 ahead of second-placed Haaland (27). His shots-on-target rate (56.3%) is the highest among the top scorers.

Bundesliga is the most attacking league

2.249 avg goals per player — the highest of all five leagues. Serie A is the most defensive at 1.750 avg goals per player.

Peak performance age is 26–28

Average goal contributions peak at 26–28 (2.427 goals per player). Players under 22 average 1.36–1.80, confirming they are still in development.

Bayern Munich dominate the +/- rankings

Seven of the top 10 players by +/- are from Bayern Munich, led by Kane (+72, +2.73/90). This shows the importance of team context — individual stats alone do not show the full picture.

Elite shooters are a rare group

282 players (of 800 with 20+ shots) achieve both high SoT% AND high G/Sh simultaneously. The shooting efficiency quadrant shows this is a meaningful differentiator for top forwards.

Underplayed gems exist in every league

Several forwards have elite per-90 goal rates (top 25%) but receive fewer than 900 minutes. These are high-value transfer targets being underutilized by their current clubs.

Goalkeepers are perfectly classifiable by machine learning

100% accuracy on GK classification. GA, Saves, Save%, and CS are uniquely non-zero only for the 187 goalkeepers — the model identifies them instantly.

DF vs MF is the hardest boundary to classify

The classifier occasionally confuses Defenders with Midfielders. This reflects a genuine football reality: defensive midfielders share the same tackle and interception profiles as central defenders. Even experienced coaches debate whether some players are DMs or CBs.

Bruno Fernandes leads assists with 21 — a European record this season

Combined with Olise's 19 assists and 15 goals, these two represent the most complete attacking midfield seasons in Europe 2025/26.

Project Summary

Task	Deliverable	Status
1 — Data Collection	Dataset loaded, 102 columns documented, column glossary produced	■
2 — Data Cleaning	2,363 clean players, 0 NaN, 10 engineered features	■
3 — EDA	Goals, shooting efficiency, aging curves, GK, +/- analysis	■
4 — Visualization	20 professional charts saved to visuals/ at 150 DPI	■
5 — Predictive Model	Role classifier 76.11% + 4 position-specific regression models	■

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github.com/mohamedmamdouh342-hash/decode-labs-data-science-internship